

Worksheet 6

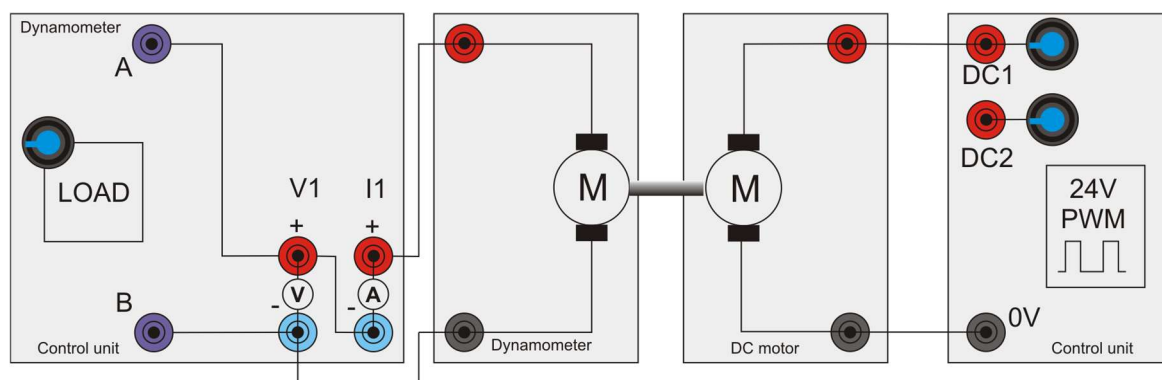
Permanent magnet generator (dynamo)

Modern electrical machines



The electrical machine in the dynamometer has exactly the same characteristics as the DC motor. The only difference is physical - the drive shaft is longer.

The photograph shows a bicycle dynamo, used to power electric lights on the front and rear of the bike.



This worksheet examines the properties of a permanent magnet generator or 'dynamo', using the dynamo found in the dynamometer.

Over to you:

- 1) Set up the system shown in the diagram.
- 2) On your PC, run the program 'Open Control_DC.bat'.
- 3) Set the dynamometer load to 80%.
- 4) Use the program to take measurements of power out, speed, and torque generated by the dynamo for DC motor driving voltages of 40%, 50%, 60%, 70%, 80% and 90%
- 5) Use the formulae in the 'Key Relationships' section on page 4 to work out the mechanical power delivered into the dynamometer.
- 6) Calculate the efficiency of the conversion from mechanical to electrical power in the dynamometer.

- 7) Use the results to plot a graph, using the axes on the next page.

You may find it easier to use a spreadsheet to do the calculations.

So what?

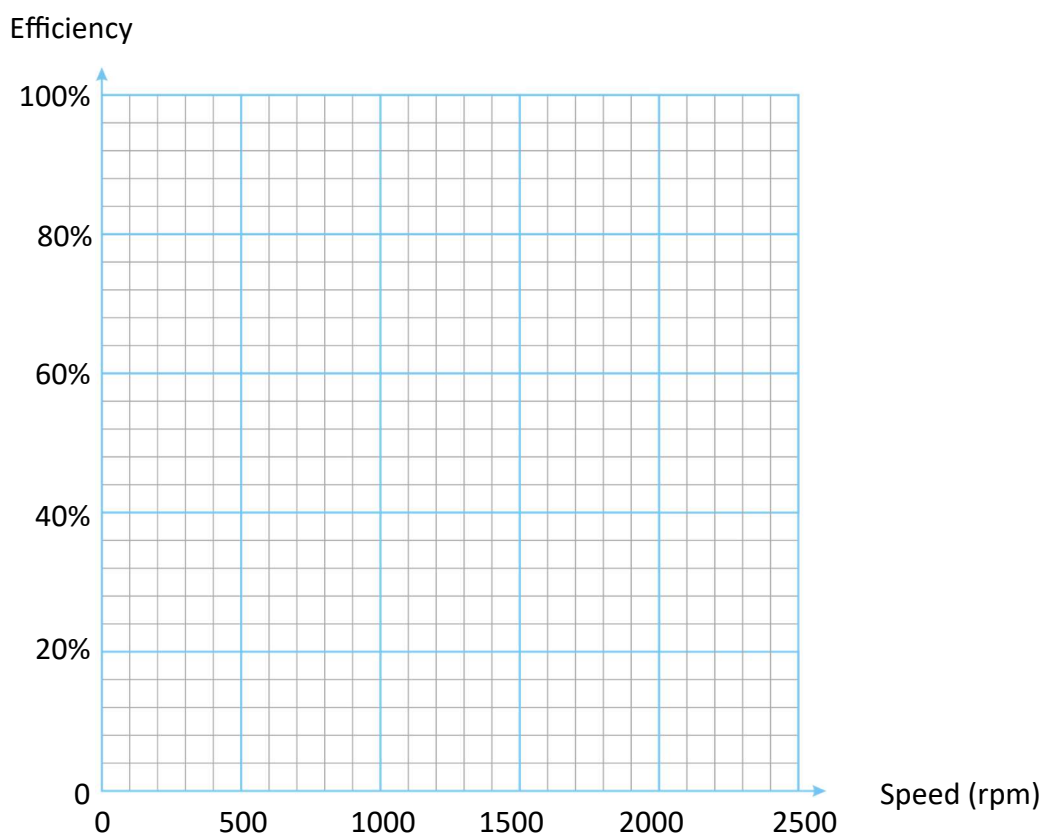
The efficiency of an electrical machine changes with its speed. Choosing the right combination of machine and gearbox for an application will help to get optimal efficiency.

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DC motor drive	Speed	Torque	Mechanical power	Dyno power	Dyno efficiency
40%					
50%					
60%					
70%					
80%					
90%					



Variation of efficiency with speed for a dynamo with a load of 80%.